

- 5 (a) Formed due to superposition/interference of the incident wave from oscillator and reflected wave from Q. 1
- The two waves have the same speed, wavelength/frequency, same amplitude and travel in opposite directions along the same line (string) 1
- (b) (i) $\frac{1}{2} \lambda = 0.30 \text{ m}$
- $\lambda = 0.60 \text{ m}$ [1] 1
- $v = f \lambda$
- $= 120 (0.60) = 72 \text{ m s}^{-1}$ [1] 1
- (ii) Within the same segment - phase difference is zero. 1

- 6 (a) Electromotive force of a source is defined as the **energy converted** 1